

CAPSTONE PROJECT

Social Network Ad Segmentation

PRESENTED BY

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GRAPHIC ERA DEEMED TO BE UNIVERSITY

B.TECH CSE(AI & DS)

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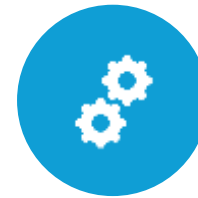
OUTLINE



Problem Statement



Proposed System/Solution



**System Development
Approach** (Technology Used)



Algorithm & Deployment



Result (Output Image)



Conclusion



Future Scope



References

PROBLEM STATEMENT

Companies struggle to effectively target the right customers for advertisements on social media platforms due to diverse user behavior and demographic differences. There's a need to segment users into meaningful groups to improve marketing strategies.



Proposed solution

- A **clustering-based approach** using **K-means** and **Agglomerative** clustering to segment social media users by age and salary, enabling better ad targeting and campaign personalization.

- **Data Collection**

- ⇒ Used social.csv
- ⇒ Selected :Age ,Estimated Salary , Gender

- **Data preprocessing**

- ⇒ Handled missing values
- ⇒ Scaled features(standardscaler)
- ⇒ Encoded categorical data

- **ML Algorithm**

- ⇒ K-means: elbow method->4 clusters
- ⇒ Agglomerative: Dendrogram->4 clusters(ward linkage)

- **Deployment**

- ⇒ visualized clusters using seaborn , matplotlib , plotly
- ⇒ Applied PCA for 2D plotting

- **Evaluation**

- ⇒ Silhouette score:0.52(moderate clustering)
- ⇒ Analyzed cluster-wise mean age/salary

- **result**

- ⇒ User grouped into 4 clear segments
- ⇒ Visual insights for targeted marketing

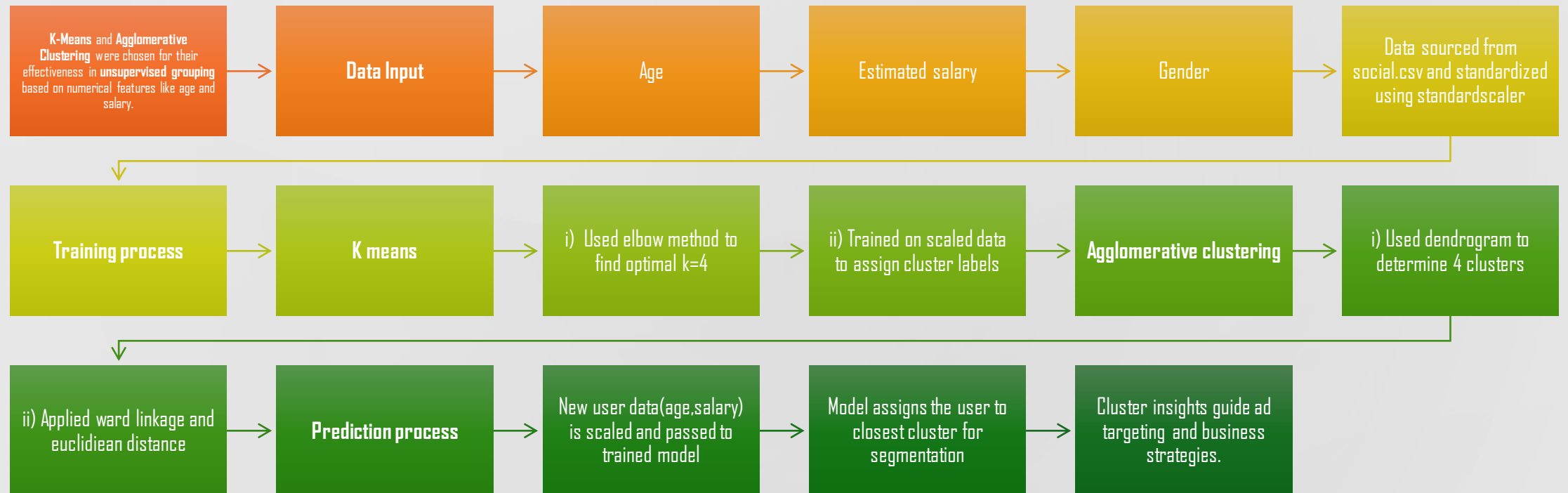


SYSTEM APPROACH

- **Language:** Python
- **Libraries:** pandas, numpy, matplotlib, seaborn, sklearn, plotly
- **Clustering Models:** K-Means, Agglomerative Clustering
- **Dimensionality Reduction:** PCA
- **Evaluation Metric:** Silhouette Score
- **Visualization:** Matplotlib, Seaborn, Plotly

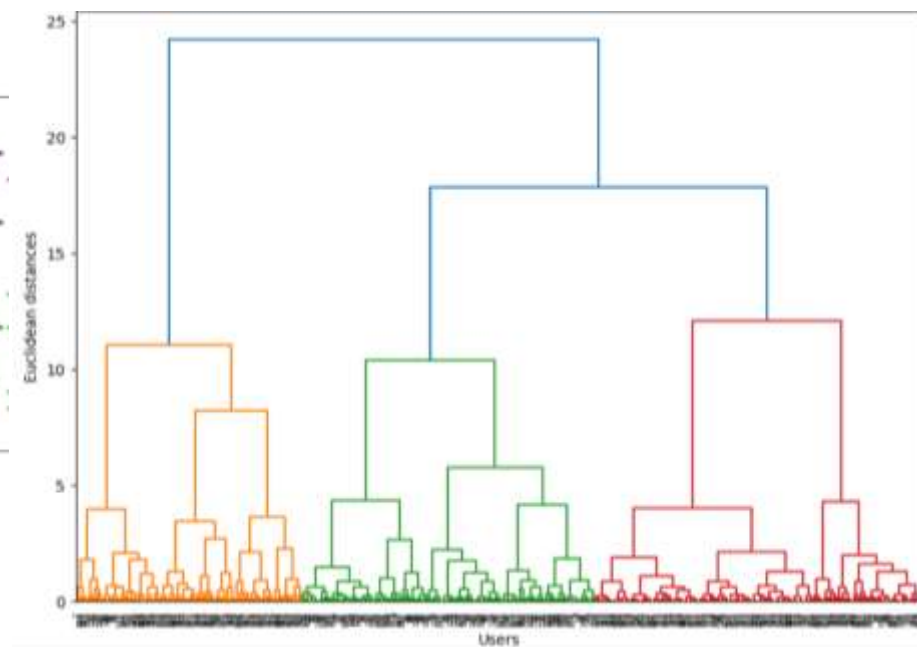
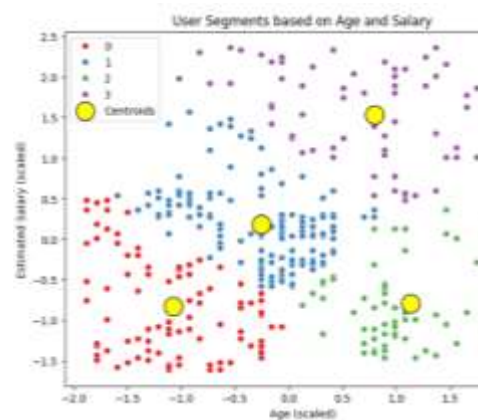
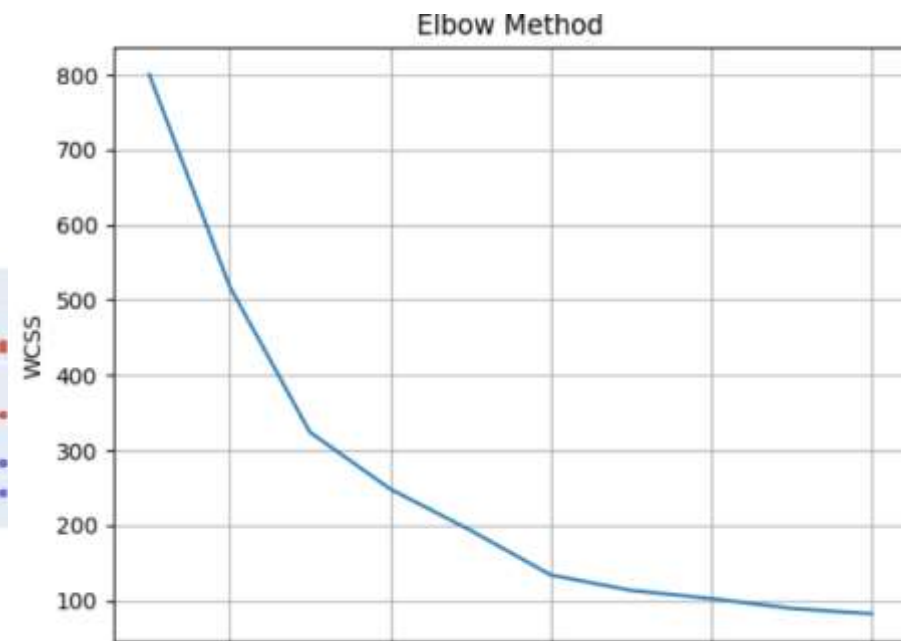
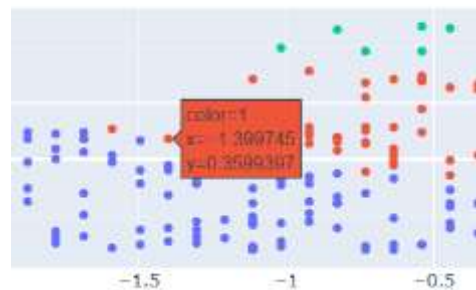


ALGORITHM & DEPLOYMENT



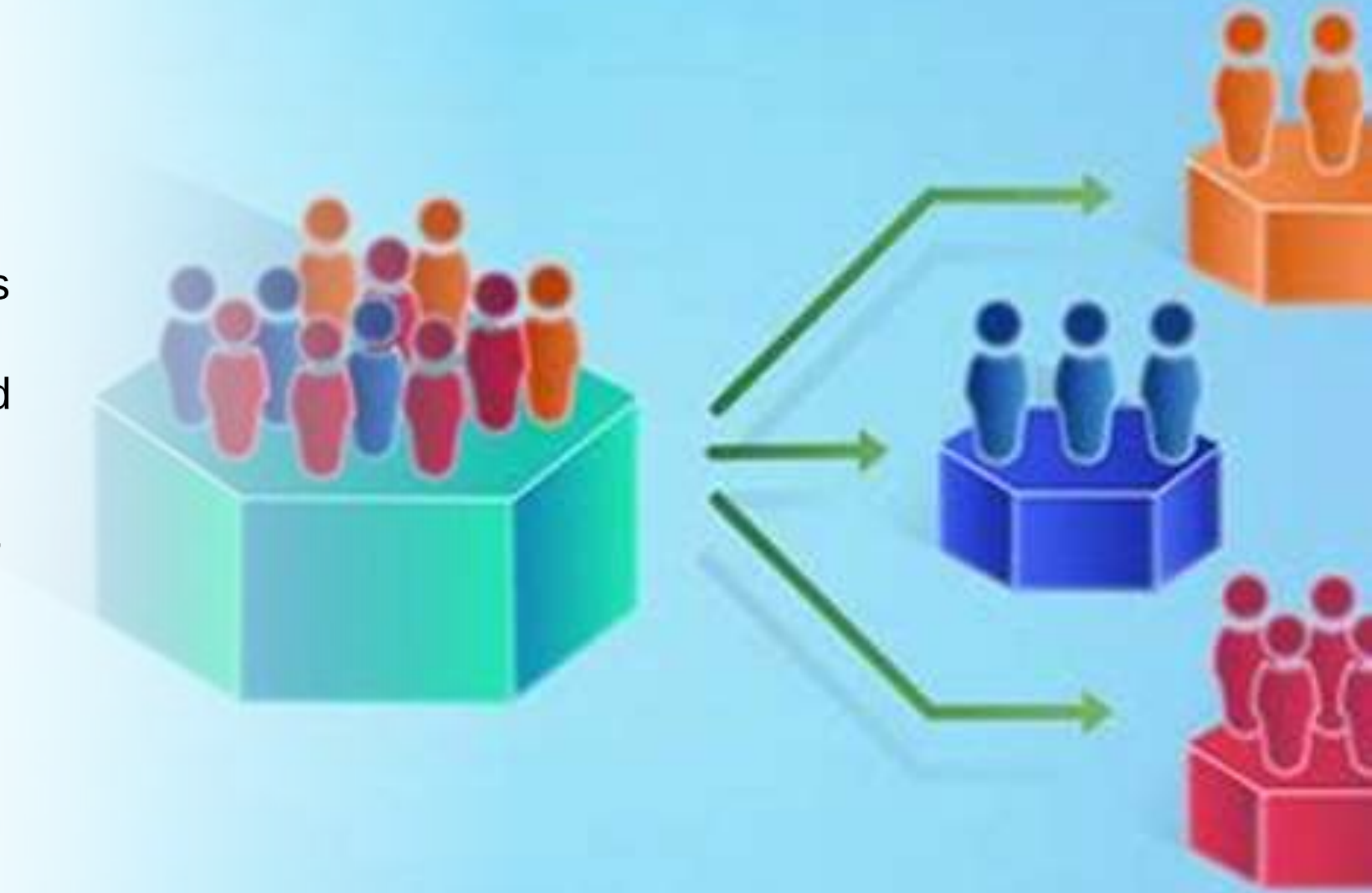
RESULT

Five User Segments



CONCLUSION

- Successfully segmented users into **4 meaningful clusters**
- Helped visualize how **Age** and **Salary** influence customer grouping
- Provided business insights for **targeted advertising** and **customer profiling**



FUTURE SCOPE

1

Add more features like
Spending Score, Gender,
Location, Activity

2

Apply clustering on real-time
data from live social
networks

3

Integrate the clustering
model with business analytics
dashboards

4

Use advanced clustering like
DBSCAN or Gaussian Mixture
Models for deeper analysis

REFERENCES

References

- Scikit-learn Documentation: <https://scikit-learn.org>
- Seaborn & Matplotlib for visualization
- Plotly Express for interactive plots
- Dataset: Custom social media dataset (social.csv)

GitHub Link: <https://github.com/ANUSHKAARORA1/Social-Network-Ad-Segmentation.git>



Thank you